

KUSHAGRA GUPTA

Machine Learning Engineer

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SUMMARY

Highly technical ML Engineer specialized in the end-to-end lifecycle of Deep Learning and Generative AI systems. Expert in architecting Agentic Orchestration pipelines using AWS Bedrock and GCP Vertex AI, and designing high-precision Computer Vision models. Proven track record in optimizing RAG using OpenSearch and ElasticSearch to bridge the gap between unstructured knowledge bases and autonomous agents. Adaptive to building scalable MLOps pipelines for automated retraining and inference on Vertex AI.

EXPERIENCE

Machine Learning Engineer

Quantiphi

Feb - 2024 Bangalore, India

- Multi-Agent AI Orchestration: Engineered an AWS-based solution utilizing Bedrock Agent core with strands to coordinate specialized Warranty, Policy, and ATA agents. Integrated S3 and OpenSearch Serverless for RAG, utilizing Lambda for autonomous JSON retrieval and SQL data synchronization to automate vehicle diagnostics in an EC2 environment.
- Engineering Excellence (Computer Vision): Architected an end-to-end UNet++ computer vision model on Vertex AI (GCP), utilizing Docker for containerized Cloud Run Functions with automated retraining and inference pipelines; reduced inference time by 30% and achieved 98% accuracy in dynamic production environments.
- Cross-Cloud Agentic Engineering: Led the architectural migration of production-grade GenAI pipelines from GCP (Cloud Run, Cloud Storage) to AWS (EC2, Lambda), utilizing MCP to orchestrate GCP services (Drive, Calendar, Chat) with a custom RAG system powered by ElasticSearch and AlloyDB; this real-time automation reduced manual intervention by 40% and was successfully adopted by multiple US universities.
- Full-Stack AI Analytics: Developed an innovative Generative AI analytics dashboard leveraging Jira APIs to provide predictive modeling for project bottlenecks, directly contributing to enterprise risk reduction.

Data Engineer

Meghplat Analytics

Nov - 2023 – Feb 2024 Internship

Python developer

R systems International

June - 2023 – Aug 2023 Internship

PROJECTS

Pneumonia Detection using SRGAN

Python, Pytorch, VGG16, SRGAN

- Enhanced the accuracy and pixel resolution of medical imaging by integrating a Super-Resolution Generative Adversarial Network (SR-GAN) with a CNN-based model, improving testing accuracy from 82% to 85.5%.

EDUCATION

SRM Institute of Science and Technology

BTech. CSE AI/ML | CGPA - 9.15/10 | 2024

Published Research Paper on increasing accuracy in Pneumonia detection using GAN technology.

CERTIFICATES

- Associate Cloud Engineer
- Professional Machine Learning Engineer

SKILLS

- Programming Languages:** Python, SQL
- Database Systems:** MySQL, Bigquery
- Data Science & Machine Learning:** Custom machine learning, GenAI, Computer vision, NLP, LSTMS, Dialogflow CX/ES, Vertex AI, Client Solution Architecture
- Cloud Technologies:** Google cloud platform (GCP)
- Libraries:** Tensorflow, PyTorch, OpenCV, Numpy, Pandas
- APIs** Jira API
- Environments** Workbench, GitLab, VScode